

PULSERETAIL

Agentic Organisation

H9CEAI Final Project · 100% CA
Customer Engagement and Artificial Intelligence

National College of Ireland · MSCAIBUS1 · 2026

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Five-Agent Pipeline

ARIA → ZARA → FORGE → LYRA → CODA

Live Prototype:

<https://ran-im.github.io/pulseretail-agents>

GitHub: <https://github.com/ran-im/pulseretail-agents>

Section 1 — Your Organisation

PulseRetail: Sustainable Activewear for Irish Fitness Enthusiasts

PulseRetail is a fictional Irish direct-to-consumer (D2C) sustainable activewear brand founded in 2022. It sells premium yoga, running, and gym apparel to 18–35 year old fitness enthusiasts in Ireland and the UK, with an average order value of €85.

The Business Challenge

PulseRetail's repeat purchase rate stands at just **18%** — less than half the industry benchmark of 35% (McKinsey, 2025). First-time buyers, acquired at a cost of €22–€28 each, are making one purchase and disappearing. Post-purchase communication consists of a single transactional confirmation email. No loyalty programme. No personalisation. No re-engagement strategy.

Why an Agentic Approach?

This challenge cannot be solved by a single AI tool. Solving it requires simultaneous expertise in market research, experience design, full-stack development, brand marketing, and strategic operations. A single generalised chatbot cannot credibly perform all of these functions at the required depth. An agentic organisation — five specialised agents, each with deep domain expertise, working in a coordinated pipeline — can move from data to strategy to working product in one coordinated cycle. That is precisely what this project demonstrates.

Section 2 — Agent Designs

The Five Agents of the PulseRetail Agentic Organisation

Each agent was built with a distinct name, personality, and system prompt anchored firmly in its archetype. No two agents overlap. Each produces output the others cannot.

ARIA — The Researcher

Superpower: Deep analysis and pattern recognition

Personality: Methodical, data-obsessed, and precise. Never speculates without evidence. Sceptical of assumptions. Her system prompt explicitly constrains her to produce a structured Research Brief: Market Context → Customer Behaviour Analysis → Root Cause Findings → Opportunity Statement → Recommended Focus Areas.

What ARIA produced: A five-section Research Brief identifying five root causes of PulseRetail's churn — no post-purchase journey, zero personalisation, absent loyalty infrastructure, passive website, no AI engagement layer — and quantifying a €2.1M annual revenue opportunity. ARIA's segment table (Yoga 34%, Running 28%, Gym 22%, Lifestyle 16%) directly informed every subsequent agent's work.

ZARA — The Designer

Superpower: Creative problem-solving and design thinking

Personality: Visionary, empathetic, boldly creative. Anchors every idea in human emotional truth. Her system prompt takes ARIA's Research Brief as input and produces a Design Specification: Solution Concept → User Journey Map → Feature Specifications → UI/UX Principles → Handoff Notes for the Maker.

What ZARA produced: The 'PulsePass' design — a five-feature loyalty intelligence portal with a 4-touchpoint customer journey (Days 3, 7, 14, 30 post-purchase), specific UI/UX principles (dark athletic aesthetic, motion as feedback, frictionless purchase path), and precise technical handoff instructions including CSS variables, animation techniques, and deployment target.

FORGE — The Maker

Superpower: Technical craftsmanship and rapid prototyping

Personality: Pragmatic, fast-moving, shipping-obsessed. Blunt about feasibility. His system prompt takes ZARA's Design Specification and produces complete HTML/CSS/JS source code deployable as a GitHub Page — zero dependencies, zero build step.

What FORGE produced: A fully functional single-page application with six interactive features: animated SVG PulseScore ring (stroke-dashoffset technique), AI recommendation product cards with hover states, 28-day engagement streak tracker, community proof strip with staggered animations, win-back offer banner (CSS pulse-glow keyframes), and a global toast notification system. Mobile-responsive. Deployed live at:

<https://ran-im.github.io/pulseretail-agents>

LYRA — The Communicator

Superpower: Persuasion and storytelling

Personality: Magnetic, persuasive, culturally attuned. Crafts stories that move people. Her system prompt takes FORGE's build and produces a complete Go-to-Market Strategy: GTM Plan → Brand Messaging Framework → Email Sequences → Social Media Plan → Launch Timeline.

What LYRA produced: Campaign 'We Missed You, Champion' — a three-phase launch strategy, segment-specific brand messaging for four customer groups (Yoga, Running, Gym, Lapsed), three complete email sequences (Day-3 'Your order arrives tomorrow', Day-14 'You're 60 points from €10 off', Day-30 'We saved something for you'), Instagram content calendar, and micro-influencer brief for Irish fitness creators.

CODA — The Manager

Superpower: Leadership and orchestration

Personality: Strategic, decisive, outcome-focused. Sees the whole board. Her system prompt reviews all four prior agents and produces an Executive Summary and Operational Plan: Summary → Strategic Alignment Assessment → Business Impact → 90-Day Roadmap → Risk Register → Board Recommendation.

What CODA produced: A full executive summary, strategic alignment rating for all four agents, revenue projection (18% → 35% repeat rate = +€104,040 annual revenue, 18:1 ROI on a €5,000 investment), a 90-day operational roadmap across three months, risk register with five identified risks and mitigations, and a formal board recommendation to deploy.

Section 3 — The Pipeline in Action

How Work Flows Through the Organisation

The pipeline is strictly sequential. No agent begins without its predecessor's output. This is not five chatbots answering five independent prompts — it is a chain of structural dependency where each output constrains and informs the next.

Handoff	What Was Passed	Evidence of Dependency
ARIA → ZARA	Five root causes + segment breakdown	ZARA's 5 features map 1:1 to ARIA's 5 recommended focus areas
ZARA → FORGE	Feature specs + CSS variables + animation brief	FORGE used --pulse-green, stroke-dashoffset, staggered setTimeout exactly as specified
FORGE → LYRA	Working portal with live data (340 pts, Emma)	LYRA's emails quote '340 points', '60 pts from €10' — grounded in FORGE's prototype
LYRA → CODA	GTM strategy + three email sequences + timeline	CODA's 90-day roadmap mirrors LYRA's three-phase launch; risk #1 cites email CTR

Working Prototype

The prototype was built by FORGE following ZARA's exact Design Specification and deployed to GitHub Pages. It demonstrates the complete PulsePass loyalty portal in action:

- Animated PulseScore ring — SVG stroke-dashoffset technique, numeric counter, progress bar to next tier
- AI Recommendation Panel — three personalised product cards (yoga category affinity) with hover states and Add-to-Bag
- Engagement Streak Tracker — 28-day calendar grid with three states (inactive / active / purchase)
- Community Proof Strip — staggered slide-in animations simulating live member activity
- Win-Back Offer Banner — pulsing CSS keyframe animation drawing attention to double-points offer

- Toast Notification System — global feedback on every user action

Live Prototype: <https://ran-im.github.io/pulseretail-agents>

Section 4 — Regulatory and Ethical Considerations

GDPR Implications

PulsePass processes personal data — purchase history, browsing behaviour, segment classification, and loyalty points — to deliver personalised recommendations and re-engagement communications. Under GDPR (Regulation (EU) 2016/679), this requires:

Lawful basis: Legitimate interest (post-purchase engagement) or explicit consent for marketing emails, clearly captured at checkout.

Transparency: Every AI recommendation surfaces a 'Why am I seeing this?' link (specified in ZARA's design). Customers must know when automated profiling influences what they are shown.

Data minimisation: Only the data required for personalisation is collected — category affinity and purchase history. No sensitive data categories are involved.

Right to erasure: Customers can request deletion of their loyalty profile at any time. The portal must include a clear data deletion pathway.

EU AI Act Considerations

PulseRetail's Re-Engagement Engine is a **low-risk AI system** under the EU AI Act (2024). It uses AI for product recommendation and engagement timing — neither of which qualifies as high-risk under Annex III (healthcare, employment, credit decisions). However, transparency obligations apply: customers should know they are interacting with AI-driven recommendations, not human curation.

Trust and Customer Perception

For the customer, trust hinges on one question: does this feel helpful or manipulative? PulsePass is designed to feel like a reward — points earned, recommendations earned, a community found. The win-back banner is honest about the offer. The email sequences are personal without being intrusive. The double-points incentive is genuine, not fabricated urgency. Provided PulseRetail honours rewards and keeps data use transparent, customers will feel seen — not surveilled.

Section 5 — Reflection and Insight

What Worked

The pipeline architecture worked better than expected. The most striking outcome was how much richer each agent's output became because of the preceding agent's work. LYRA's email sequences are specific and credible — 'You're 60 points from your €10 reward' — because FORGE had already built a portal where 340 points was the customer state. A single agent asked to write marketing emails could

not have produced that specificity. The handoff created emergent depth that no individual agent could achieve alone.

The separation of personality and domain expertise also worked well. ARIA's clinical, data-driven voice is genuinely distinct from ZARA's visionary tone, which is distinct from FORGE's blunt engineer voice. These are not the same model producing five slightly different responses — they are different professional identities producing fundamentally different types of work product.

What Didn't Work

The prototype requires a real backend to become production-ready. FORGE correctly flagged this: Emma's 340 points are hardcoded. In production, every customer would need live data from Shopify and a recommendation engine consuming real purchase history. The prototype is a high-fidelity demonstration, not a live system. CODA's risk register also identified that low portal adoption — customers not clicking from emails to the portal — is the highest probability risk. The pipeline produced a beautiful portal, but if the Day-3 email subject line is wrong, no one will ever see it.

What I Learned About Multi-Agent Collaboration

The handoff is the hardest part — and the most important. An agent that produces clean, structured, explicitly-labelled output (ARIA's numbered root causes, ZARA's bullet-pointed handoff notes) creates far more usable input for the next agent than prose. Structured outputs are the connective tissue of multi-agent systems.

The five archetypes reflect a genuine truth about how organisations work. Research, design, engineering, marketing, and management are not interchangeable. Each has a distinct epistemology. The best outcome comes from five genuine specialists, not one generalist wearing five hats.

If I Had More Time

I would build the Shopify integration, replace hardcoded data with real personalisation, and run the Day-3 email sequence through a live A/B test. I would also explore whether CODA could monitor post-launch performance data and trigger ARIA to re-run the analysis cycle — creating a self-improving feedback loop rather than a one-shot pipeline.

AI Usage Declaration

Generative AI use is required for this assignment. All AI-generated content was reviewed and edited for accuracy. System prompt design, agent personality definition, orchestration logic, and critical evaluation are the student's original contribution.

Agent	Model	Role in Pipeline
ARIA	Claude Sonnet 4.6	Customer Intelligence Analyst — Research Brief
ZARA	Claude Sonnet 4.6	Experience Design Architect — Design Specification
FORGE	Claude Sonnet 4.6	Product Engineer — Working Prototype (index.html)
LYRA	Claude Sonnet 4.6	Brand & Marketing Strategist — GTM Strategy

Agent	Model	Role in Pipeline
CODA	Claude Sonnet 4.6	Chief Operating Agent — Executive Summary

GitHub Repository: <https://github.com/ran-im/pulseretail-agents> · Live Prototype: <https://ran-im.github.io/pulseretail-agents> · Word count: ~2,100 words